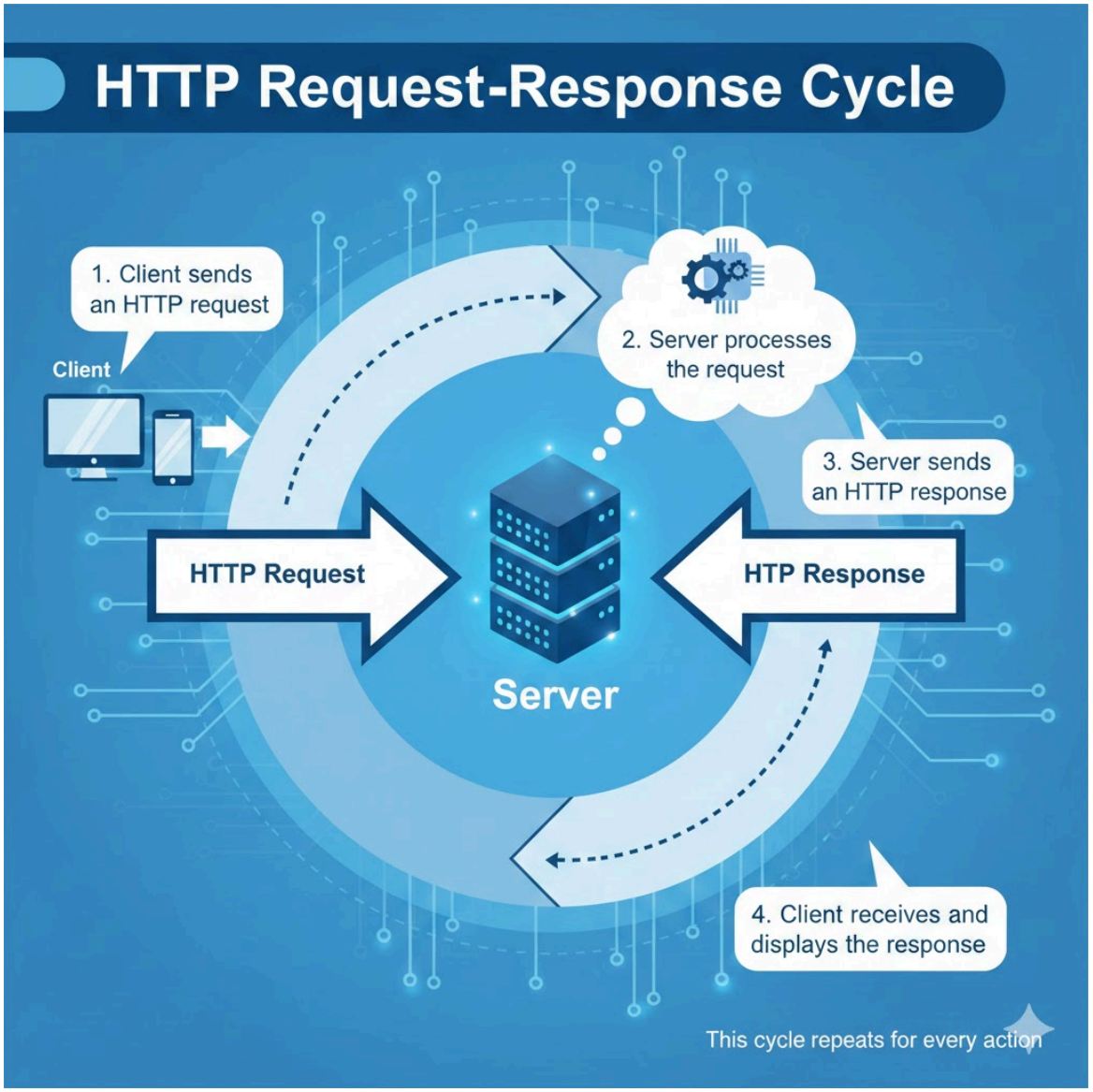
21. Understanding of HTTP

▼ HTTP Basics

- **HTTP (Hypertext Transfer Protocol)** is used to transfer data over the internet.
- It is the **foundation of the World Wide Web (WWW)**.
- HTTP works at the **application layer**.
- It acts like a **delivery system** between client and server.

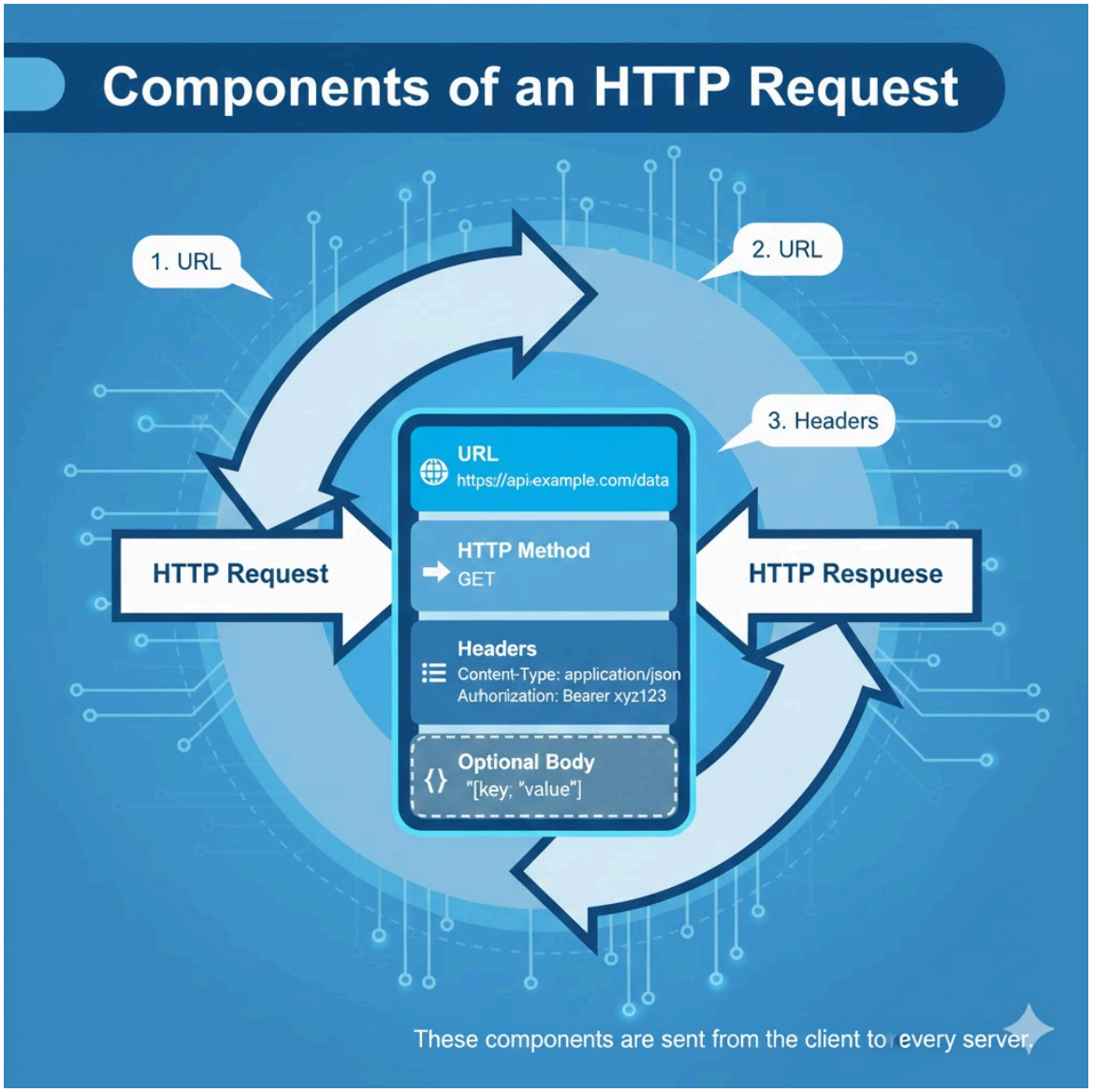
▼ HTTP Request–Response Cycle

- Client sends an **HTTP request**.
- Server **processes** the request.
- Server sends an **HTTP response**.
- Client **receives and displays** the response.
- This cycle repeats for every action.

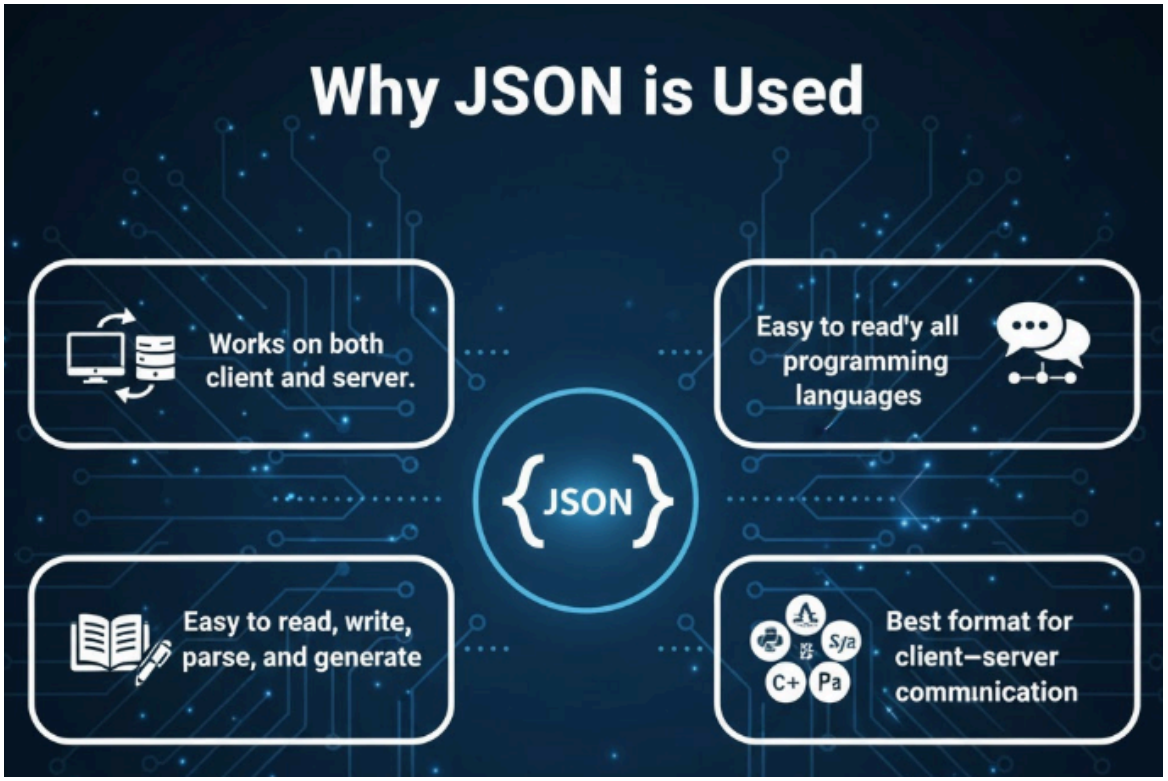


▼ Components of an HTTP Request

- **URL**
- **HTTP Method**
- **Headers**
- **Optional Body**

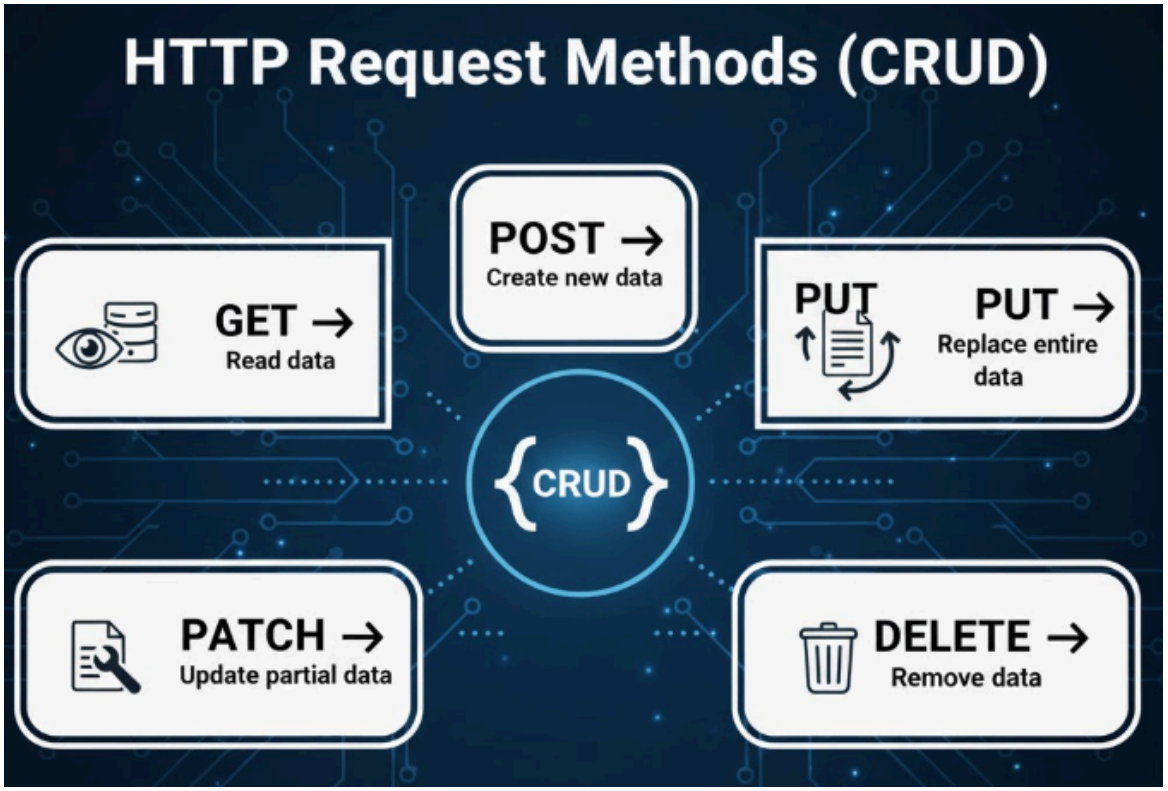


▼ Why JSON is Used?



▼ HTTP Request Methods (CRUD)

1. **GET** → Read data
2. **POST** → Create new data
3. **PUT** → Replace entire data
4. **PATCH** → Update partial data
5. **DELETE** → Remove data



▼ 1. `GET` Method

- Used to **retrieve data**.
- Does **not modify** data

▼ 2. `POST` Method

- Used to **create new resources**.
- Sends data in **request body**.

▼ 3. `PUT` Method

- **Replaces the entire resource**.
- Missing fields are deleted.

▼ 4. `PATCH` Method

- **Updates only specific fields**.
- Other data remains unchanged.
- More **efficient and flexible** than PUT.

- Preferred for updates in frontend apps.

▼ 5. `DELETE` Method

- Used to **delete a resource**.
- Usually requires an ID.

▼ HTTP Headers

- Sent as **key–value pairs**.
- Provide extra information (content type, authorization, etc.).
- Used in both request and response.

▼ HTTP Request Body

- Contains data sent from client to server.
- Used mainly with **POST, PUT, PATCH**.

▼ HTTP Response

- Sent by server after processing request.
- Contains:
  - Status line
  - Headers
  - Response body (JSON, HTML, etc.)

▼ HTTP Status Code Categories

- **100–199** → Informational Status Code
- **200–299** → Success Status Code
- **300–399** → Redirectional Status Code
- **400–499** → Client Error Status Code
- **500–599** → Server Error Status Code

▼ Commonly Used Status Codes

- **200** → OK
- **201** → Created
- **204** → No Content
- **400** → Bad Request
- **401** → Unauthorized
- **403** → Forbidden
- **404** → Not Found
- **500** → Internal Server Error